Day 7: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

C1-3

How many three-digit positive integers have an even last digit and no repeated digits?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C2-3

A committee of 3 is chosen from 5 women and 4 men. How many committees contain at least one woman?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C3-3

How many integers from 1 through 100 are divisible by 3 or 5?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C1-2

A code has one letter from A,B,C followed by two digits from 0,…,9. Repetition is allowed. How many codes exist?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C2-2

From 8 distinct students, how many groups of 3 can be chosen? How many ways can president, secretary and treasurer be assigned to different students?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C3-2

In a class of 30, 18 study French and 15 study German; 7 study both. How many study at least one, and how many neither?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C2-1

Explain the difference between choosing r people from n and assigning r distinct ordered positions to n people. State both counts for $0 \le r \le n$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C3-1

State the formula for the size of the union of two finite sets A and B. Why is there a subtraction?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____